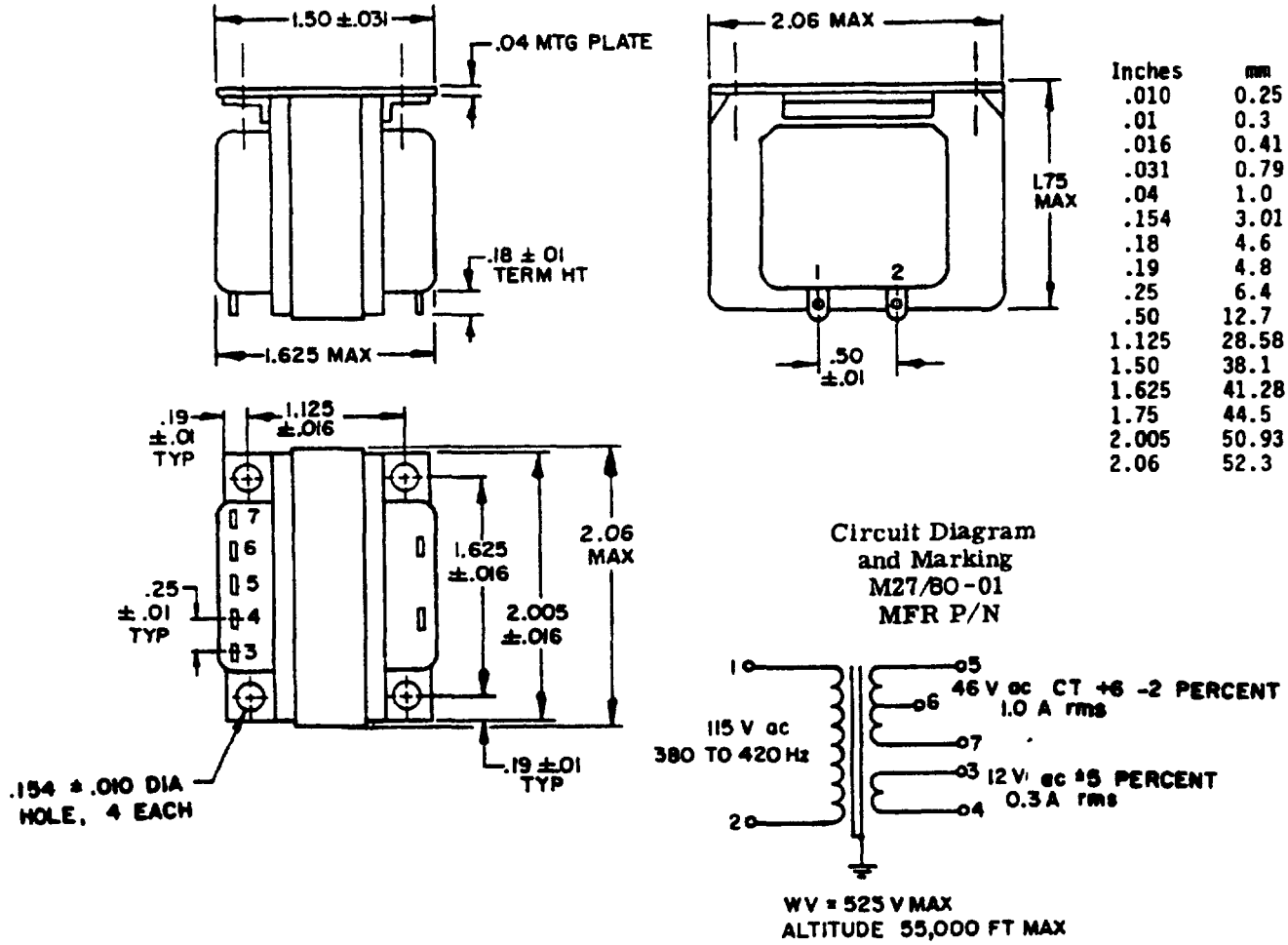
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEPDOWN, TF5R03ZZ897

Ⓐ THIS SPECIFICATION IS INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformers described herein shall consist of this specification and the latest issue of MIL-T-27.



Inches	mm
.010	0.25
.01	0.3
.016	0.41
.031	0.79
.04	1.0
.154	3.01
.18	4.6
.19	4.8
.25	6.4
.50	12.7
1.125	28.58
1.50	38.1
1.625	41.28
1.75	44.5
2.005	50.93
2.06	52.3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.015 (.40 mm).
4. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configuration.

Ⓐ denotes change

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-2): 115 V, 400 Hz 5* percent.

Secondary voltage:

(3-4): 12 V ac ±5 percent, 0.3 ampere rms.

(5-7): 46 V ac ct +6 percent, -2 percent, 1.0 ampere rms.

Working voltage:

(1-2): 525 V maximum.

(3-4) (5-7): 525 V maximum.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Altitude: 55,000 feet, maximum.

Terminals: Solder type, brass, hot tinned dip.

Terminal height: 0.18 inch (4.6 mm).

Weight: 1 pound.

Operating temperature range: -55°C to 105°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 1,500 V rms.

At reduced barometric pressure: 669 V rms.

Electrical characteristics:

No load: With 115 V, 400 Hz across (1-2), current in (1-2) shall not exceed 0.75 mA and the power in (1-2) shall not exceed 4 watts.

Voltage across (3-4): 12.38 V ±0.10; (5-7): 48.60 V ±0.60.

Rated load: With 115 V, 400 Hz across (1-2), the voltage across (3-4) shall be 12 V ±5 percent; voltage across (5-7) shall be 46 V ct +6 percent, -2 percent.

DC resistance:

(1-2): 4.5 ohms ±20 percent.

(3-4): 0.55 ohm ±20 percent.

(5-7): 0.94 ohm ±20 percent.

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Polarity: Additive, with terminals 2 and 3 and 4 and 5 connected.

Temperature rise: 40°C maximum at an ambient temperature of 65°C maximum.

Marking location: See figure 1.

Part number: M27/80-01.

Ⓐ Quality assurance provisions:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A and B tests of MIL-T-27 shall be applicable.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0671)

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